

Artificial Intelligence in Mobile Oral Cancer Screening: A Survey of Current Applications and Technologies

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Abstract—Oral cancer remains a significant global burden due to many cases being diagnosed too late. The combination of easily accessible mobile phones and artificial intelligence techniques for image analysis opens promising perspectives regarding the possibility of an inexpensive and easy point-of-care test even in areas where specialized healthcare institutions are absent. The current paper provides a review of four mobile AI-based diagnostic applications (BerryCare, OrCA, Aarogya Aarohan, and MeMoSA) and analyzes the features related to their structure, input/output operations, AI algorithms involved, and application in medical practice. Some common tradeoffs typical for AI-based oral cancer diagnostic software include the conflict between achieving high classification accuracy and real-time requirements, between providing an established protocol for imaging and practical constraints, and between providing an algorithmic diagnosis and a diagnosis made by a clinician using the app. Although initial results demonstrate some promise, clinical evidence published in peer-reviewed journals is yet to follow, as well as some improvements in the interpretability of AI models, data privacy management, and regulatory frameworks.

Index Terms—Oral cancer, mobile health, artificial intelligence, deep learning, screening, early detection, convolutional neural networks, medical imaging

I. INTRODUCTION

Oral cancer accounts for roughly 5% of cancers worldwide; in India, that figure is approximately 40% of all cancer cases, with close to 60,000 new diagnoses reported annually [1]. Despite decades of clinical attention, survival rates remain poor—not because of inadequate treatment, but because of persistent late-stage detection.

The prognostic implications of early diagnosis are clear. Five-year survival for stage I oral cancer exceeds 80%, while advanced-stage disease carries survival rates below 40% [5]. Conventional screening depends on trained specialists within clinical infrastructure that is unavailable across much of rural India and comparable low-resource settings [6]. Mobile health (mHealth) platforms have emerged as one response, using smartphone cameras alongside AI-based image analysis to enable point-of-care screening without specialist presence.

Deep learning, particularly CNN-based architectures and more recently vision transformers, has reached near-specialist

performance across several medical imaging domains [7], [8]. Applied to smartphone-captured oral cavity images, these techniques offer a low-cost and potentially scalable approach to triage [9]. Whether they translate reliably into field conditions remains an open question that this survey examines.

Four platforms are analyzed here: BerryCare, OrCA, Aarogya Aarohan, and MeMoSA. Each reflects a different design philosophy. The analysis covers technical architecture, input-output mechanisms, AI methodology, validation status, and deployment context. Shared limitations across platforms receive particular attention, since these may indicate structural rather than platform-specific problems. The primary contributions are:

- A review of current mobile AI screening technologies and their technical architectures
- Comparative analysis of input/output mechanisms and AI methodologies across platforms
- Evaluation of performance characteristics, validation status, and associated challenges
- Discussion of technical and clinical gaps requiring future research attention

II. BACKGROUND

A. Oral Cancer and its Clinical Landscape

Oral cancer includes malignancies of the lip, tongue, floor of the mouth, buccal mucosa, gingiva, hard palate, and oropharynx [5]. Primary risk factors include tobacco use, alcohol consumption, betel quid chewing, and HPV infection; risk increases substantially when these co-occur [10].

Oral potentially malignant disorders (OPMDs) which include leukoplakia, erythroplakia, and oral submucous fibrosis, often precede frank malignancy [11]. These lesions represent the most actionable intervention window, so their reliable detection is central to any screening strategy. Standard clinical screening involves visual inspection of the oral cavity for color changes, texture abnormalities, or persistent ulceration [12]. This requires trained examiners and adequate lighting, conditions that are not guaranteed in community or field settings.

B. Deep Learning for Medical Imaging

The use of deep learning in medical image processing has changed considerably by shifting from hand-crafted features to learning through the use of labeled images [8]. CNN architectures, including ResNet, VGG, and EfficientNet, have achieved expert accuracy in areas such as skin lesion classification [7], diabetic retinopathy [14], and computational pathology [15].

In contrast to these CNN models, vision transformers (ViT) avoid using convolutional biases to detect global spatial structures in an image via self-attention [16]. This approach is crucial when considering how to accurately detect oral lesions since the surroundings might play an essential role in the diagnostic process. The MobileViT model is more feasible for implementation due to its reduced architecture that preserves the ability to extract global features in a device with restricted memory capacity [17].

Transfer learning techniques, in which the model is initialized with the pre-trained weights from natural image data and then finetuned to learn from domain-specific images, have yielded promising results in cases where data are limited [18]. Transfer learning is used in some fashion by all four architectures described here. However, this technique has certain limitations as the domain discrepancy between ImageNet and intraoral photography might lead to unexpected results not detected by any validation dataset.

C. Mobile Health for Oral Cancer Screening

The case for mHealth-based oral cancer screening rests on well-documented accessibility arguments: reduced infrastructure requirements, real-time decision support, and facilitation of specialist referral from remote settings [6]. Proof-of-concept studies have demonstrated technical feasibility of smartphone-based oral lesion detection [9], though feasibility under controlled conditions does not straightforwardly generalize to performance across diverse field settings, user populations, and imaging hardware [19]. The gap between demonstrated accuracy in published studies and practical clinical utility remains the central unresolved question in this domain.

III. METHODOLOGY

PubMed, IEEE Xplore, and Google Scholar searches yielded platforms that used terms like oral cancer, mobile screening, smartphone, artificial intelligence, deep learning, and mHealth. The following criteria for inclusion were applied when selecting platforms: operates in a smartphone; uses an algorithm based on artificial intelligence or machine learning to analyze data; focuses on detecting oral cancer/OPMD; offers documentation of any kind. Telemedicine platforms which lacked the AI capability and research platforms without actual implementation have not been included into the review.

The information on the technological basis of each platform, signals it uses as an input, algorithms for AI analysis, its output visualization, clinicians' involvement, performed validation and platform deployment was collected during the analysis. The algorithm utilized by the platform is usually presented

as proprietary software, thus making analysis dependent on public information only.

IV. CURRENT MOBILE AI SCREENING PLATFORMS

A. BerryCare

BerryCare qualifies as a SaMD, putting it under the scope of regulations applicable to clinical software [20]. The key functionality of BerryCare is automated risk assessment of the oral cavity via images captured on a smartphone.

1) *Input Mechanism*: Information acquired by BerryCare includes demographic data (patient's name, age, sex, geographic location), behavioral risk factors (use of cigarettes, alcohol, tobacco, including duration), presence of symptoms (oral pain), and images of the oral cavity from different viewing angles, including lips, inner surface of cheeks, tongue, and bottom of the mouth.

2) *AI Methodology*: Model architecture is proprietary and unknown; it is known only that deep learning methods are used for analysis. No details about network architecture, training dataset, sample balance, or cross-validation are publicly available, making independent verification of claimed results impossible.

3) *Output and Clinical Interface*: Assessment results are provided to the user via smartphone interface, which displays binary risk classification ("suspicious" or "non-suspicious"), as well as visual notifications, behavioral risk factors, certainty measure, and advice about next steps.

4) *Regulatory and Deployment Status*: Regulatory registration of SaMD involves certain level of oversight. However, it is not clear which jurisdictions and what type of clinical validation (if any) are covered by it, as there are no relevant papers in scientific literature.

B. OrCA – Oral Cancer Assessment

OrCA operates within an alternative design space compared to the other platforms, focusing mainly on clinical documentation and workflow with a relatively limited application of AI [21].

1) *Input Mechanism*: Demographics, behavioral risk factors, a medical history of hypertension, cardiac diseases, respiratory diseases, and diabetes, intraoral and extraoral high-resolution images are captured with OrCA. These images may be acquired using compatible imaging devices with uniform lighting and up to 16× magnification — this significantly exceeds ordinary smartphone camera capabilities. A guided palpatory exam process and protocol are provided as well.

2) *AI Methodology*: There is no technical information about AI architecture and training used in OrCA. The application assumes that the process of analysis is possible but clearly presents itself as a clinical support system rather than an independent AI. In our opinion, this decision may indicate conscious awareness of premature automation given validation experience at this point.

3) *Output and Clinical Interface*: As a result, structured clinical reports are generated in OrCA, containing images, observations made during the process, findings of lesions and other clinical notes. Teleconsultation and referrals are the use cases suggested by the developers for these reports; their cloud-based storage and sharing capabilities are provided.

C. Aarogya Aarohan

Aarogya Aarohan was created by a consortium including Indian Institute of Science and AIIMS Delhi, which makes it one of the more institutionalized platforms under review [22]. The key characteristic of the solution from a technical perspective is on-device AI inference based on vision transformers.

1) *Input Mechanism*: Users are guided to take 8–9 images of the patient’s oral cavity using a flashlight in a clockwise manner. Thus, there is an effort to define the range of the inspection as precisely as possible and limit operator variance. The second part of the input is clinical habits data, structured symptom questionnaire, and demographic details.

2) *AI Methodology*: Binary classification (“suspicious” or “non-suspicious”) is done on the device with MobileViTv2, a compact vision transformer architecture optimized for deployment on the edge [17]. Inference is performed on-device and can be done offline as well, which greatly facilitates usage in a rural setting. The model evaluates images taken with a flashlight — that is, white-light images. Sensitivity and specificity are claimed to be approximately 85%, which sounds like a significant number; however, exact circumstances under which this claim is true remain unclear.

Global spatial context modeling is possible thanks to MobileViTv2’s architecture, thereby capturing the interplay between lesions and surrounding oral tissue. However, the potential drawback is a higher degree of training complexity compared to common CNN approaches, along with some sensitivity to image noise in different field environments.

3) *Output and Clinical Interface*: Risk classification based on confidence scores is presented directly in the user application. Screening data and images are then uploaded to a cloud specialist dashboard for remote review and initial diagnosis. The algorithm produces referral recommendations stratified according to risk levels: monitoring, counseling, or clinical assessment.

4) *Technical Challenges*: Critical limitations are: (1) the need for extensive datasets of high-quality annotated images for training the transformer, which is currently lacking; (2) the problem of class imbalance; (3) inconsistency of performance between different smartphone models; and (4) variance between field image quality and training data distribution leading to uncertain results.

D. MeMoSA – Mobile Mouth Screening Anywhere

MeMoSA is the best documented of the four programs, with peer-reviewed articles describing its architecture and performance [23], [24]. In addition to screening images, it performs localization of lesions—a much more challenging task.

1) *Input Mechanism*: The input comprises smartphone-acquired photographs of the tongue, buccal mucosa, and floor of the mouth. Patient demographic information such as age, gender, and risk factors are added if available, although the degree of effect of the clinical context on the model output is not discussed

2) *AI Methodology*: MeMoSA uses a dual approach of deep learning algorithms.

Classification Module: ResNet-101 is fine-tuned from ImageNet using transfer learning and used to classify images on three levels: no lesion, low risk, or high risk/malignancy [13].

Detection Module: A Faster R-CNN detector is trained using ResNet-101 architecture for lesion localization and creation of bounding boxes with class labels and confidence scores [25]. Training samples are clinicians’ oral images annotated using composite annotation [23].

Using ResNet-101 and Faster R-CNN allows excellent feature extraction and spatial localization capabilities, albeit at relatively high computation costs. Transfer learning from natural images to intraoral images poses a domain discrepancy problem.

3) *Output and Clinical Interface*: MeMoSA generates multiple outputs, including image-level risk category, lesion location through bounding boxes, and the application-level recommendation on whether or not the patient needs referral. In some implementations, MeMoSA allows for sending images to be reviewed asynchronously by specialists along with annotated feedback provided back to the users.

4) *Performance Evaluation*: According to publicly available performance studies, the tool shows very good metrics of precision, recall, and F1 in lesion presence prediction; however, referral support has somewhat weaker results [24]. Localizing smaller lesions, areas with poor contrast, or inconsistent light conditions makes Faster R-CNN less reliable for this use case. This is not particularly surprising due to the architecture’s nature, yet still important from a medical point of view.

V. COMPARATIVE ANALYSIS AND DISCUSSION

A. Technical Approaches

The four platforms differ substantially in their computational philosophy, reflecting different assessments of the deployment problem (Table I).

Architecture Choices: MeMoSA and Aarogya Aarohan represent opposing ends of the accuracy-efficiency spectrum. MeMoSA prioritizes detection fidelity and spatial localization using ResNet-101 and Faster R-CNN—computationally intensive architectures suited to settings with reliable connectivity and processing resources. Aarogya Aarohan’s MobileViTv2 makes the opposite trade, optimizing for offline on-device inference at acceptable classification accuracy. Given that the target deployment context for both platforms includes rural, low-connectivity settings, Aarogya Aarohan’s architectural choice appears better calibrated to actual field constraints—though whether the ~85% sensitivity holds under those conditions has not been rigorously demonstrated.

TABLE I
COMPARATIVE SUMMARY OF MOBILE AI ORAL CANCER SCREENING
PLATFORMS

Platform	AI Method	Key Feature	Deployment
BerryCare	Undisclosed DL	SaMD registered	Commercial
OrCA	Limited AI	Clinical workflow	Pilot
Aarogya Aarohan	MobileViTv2	On-device inference	Field testing
MeMoSA	ResNet-101, Faster R-CNN	Lesion localization	Research

Image Acquisition: Standardization approaches vary considerably. Aarogya Aarohan’s clockwise 8–9 image protocol attempts to enforce anatomical coverage and reduce user variability. OrCA’s controlled illumination and magnification support more consistent image quality in clinical settings. BerryCare and MeMoSA rely on less constrained smartphone photography, accepting greater variability in exchange for simpler deployment. Image quality is consistently among the strongest predictors of model accuracy in oral lesion classification; the absence of standardization likely contributes to performance variability under field conditions.

Clinical Integration: The OrCA focus on being a workflow tool, as opposed to an autonomous classifier, should be taken into account. OrCA essentially accepts the possibility that the use of AI to diagnose oral lesions autonomously is perhaps premature at this point—an assertion that seems well-justified based on the paucity of available validation data.

B. Common Strengths

Several advantages cut across all reviewed platforms.

Accessibility: Smartphone-based screening reduces infrastructure requirements relative to clinic-based approaches. By enabling non-specialist deployment, these platforms could identify suspicious cases earlier in disease progression, when treatment outcomes are substantially better.

Decision Support: Automated risk stratification and referral guidance support systematic triage where specialist training is limited. The structured output formats may reduce variability in referral decisions compared to unassisted community health worker assessments.

Data Management: Digital image capture and cloud synchronization open possibilities for longitudinal monitoring and population-level surveillance that paper-based systems cannot support. This secondary benefit may prove as clinically valuable as the primary screening function over time.

C. Common Limitations

Validation Gap: The most important limitation is the scarcity of prospective, peer-reviewed clinical validation. Published performance figures, where available, largely reflect retrospective evaluations on curated datasets. Sensitivity and specificity under real-world field conditions, with heterogeneous imaging hardware, variable operator skill, and diverse

patient populations, remain insufficiently characterized [19]. MeMoSA is the partial exception, with peer-reviewed publications, but even these do not comprehensively address prospective field performance.

Image Quality Dependency: All systems are sensitive to lighting, focus, angle, and field of view. Standardized capture protocols and operator training can partially mitigate this, but neither is easy to implement consistently at scale across diverse deployment settings [6].

Model Opacity: Deep learning models do not readily expose their reasoning. None of the reviewed platforms incorporate explainability mechanisms allowing clinicians to understand which image features drove a particular classification. This limits clinical trust and makes error diagnosis difficult [26].

False Negative Risk: In screening contexts, false negatives carry particular clinical weight, as they may provide false reassurance to patients with actionable disease. The sensitivity-specificity trade-off is inadequately documented for any of these platforms under field conditions [9].

Data Privacy: These platforms collect sensitive health data and images from potentially vulnerable populations. Recording of data handling procedures, encryption methods, data retention periods, and data-sharing procedures with other parties is inadequate on all four platforms, an issue that carries ethical and regulatory implications.

Complementary Role: All four are diagnostic aids, but no platform substitutes for biopsy and histopathology in making an exact diagnosis. It is not clear whether this limitation in scope is properly communicated to community health workers using the platforms.

VI. CHALLENGES AND FUTURE DIRECTIONS

A. Technical Challenges

1) *Dataset Issues:* Datasets for training oral cancer detection models continue to be rare, heterogeneous, and generally unbalanced, with an overwhelming number of normal cases compared to abnormal cases. The quality of annotations is affected by the expertise of annotators, and currently available datasets do not sufficiently reflect population diversity, lesion characteristics, and imaging circumstances [23].

2) *Model Explainability:* The interpretation of models can be facilitated using gradient-based techniques like Grad-CAM [27] and attention analysis, although their practical clinical value is still questionable. It might be necessary to employ more robust interpretability paradigms, such as concept-oriented reasoning and uncertainty assessment, to ensure that deep learning algorithms are clinically acceptable [?].

3) *Multimodal Data:* Current platforms utilize mainly white light smartphone images for classification. Clinical information, structured risk factors, and multispectral imaging techniques (autofluorescence, narrow band imaging) could further enhance the distinction between oral malignancies and benign OPMDs, especially when lesions are visually similar under white light illumination [28]. The extent to which

improvements translate to real-world conditions needs to be evaluated experimentally.

4) *Online and Federated Learning*: The deployed models must cope with distributional changes since they will be applied to diverse demographics and imaging modalities. Federated learning involves updating models based on decentralized local data without aggregating sensitive patient images [29].

B. Clinical and Deployment Challenges

1) *Standardization*: Without standardization of image acquisition techniques, quality indicators, and reporting, inter-platform comparisons and aggregations of evidence become highly unlikely. Industry-across standardized acquisition protocols similar to those established for other medical imaging modalities could be highly useful in this field [6].

2) *Clinical Validation*: Further prospective studies of mobile AI-based eye screening compared against expert evaluation in situ are clearly warranted. These reports should provide such parameters as sensitivity, specificity, positive and negative predictive value, along with implementation indicators. The evidence available currently does not allow one to draw any solid conclusions on the positioning of these instruments in the clinic [19].

3) *User Training*: Quality of the images taken, the results obtained, and decisions to refer patients are all contingent upon the operator's competence. The area of CHWs' training needs further investigation and investment, considering its potentially high impact on tool performance in reality [6].

4) *Clinical Workflow Integration*: Tools which do not integrate with the current health infrastructure, referral pathways, and specialists' routines risk yielding valuable but neglected outputs. Sustainable implementation cannot rely solely on stand-alone technologies [9].

5) *Regulatory Frameworks*: AI medical devices necessitate flexible regulatory mechanisms able to address the issues of model updating, post-market surveillance, and performance tracking in situ—mechanisms that are underdeveloped in LMICs currently [30]. BerryCare's SaMD registration is one example of a first step towards regulation, however the appropriateness of current frameworks for AI-based models remains disputable.

C. Equity and Access

It goes without saying that these tools will yield maximum benefits where specialist services are scarce. Prioritization of target populations requires addressing aspects of digital literacy, tool affordability, internet connectivity, and community readiness. Technology development without community engagement may yield problems not visible in technical performance metrics [6].

D. Future Research Directions

Among possible avenues for further development, the following come to mind:

- Development of lightweight, explainable algorithms optimized for mobile platforms

- Multi-modal fusion frameworks that consider clinical, imaging, and behavioral risks
- Federated learning approaches enabling private collaboration of distributed models
- Randomized controlled trials comparing mobile screening against standard procedures
- Health economics evaluations to assess tool cost-effectiveness in various settings
- Implementation research to address acceptability and integration issues

VII. CONCLUSION

Four mobile screening platforms based on AI technology, BerryCare, OrCA, Aarogya Aarohan, and MeMoSA, are discussed from four perspectives: technological architecture, methodological approach of AI, clinical user interface, and application context. These mobile screening platforms vary significantly in terms of their underlying philosophy, ranging from highly computation-intensive localization networks on one hand to on-device transformer-based inference on the other, complemented by a structured documentation platform occupying an intentionally distinct position.

The necessity for mobile screening based on AI technology stems from the reality that the prevalence of oral cancer in India and similar countries is underdiagnosed, while the required infrastructure to perform the screening cannot be scaled up. Nevertheless, there is a lack of a solid scientific foundation that is needed to ensure the proper use of such mobile apps in practice.

It seems sensible to view mobile AI technologies for screening as elements of diagnostic chains. Such screening devices could potentially increase the accuracy of diagnosis, broaden the range of patients that get access to specialists, and identify the cases that otherwise might never get diagnosed. Nonetheless, this potential needs to be realized by means of technological progress and clinical research.

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